

THE UNIVERSITY OF
SYDNEY

HollowHaven

Project report

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Contents

1 Project Overview **2**

2 Problem and Design Opportunity **2**

3 Design Goals **3**

4 Proposed Solution **3**

5 System Architecture **4**

6 Context-Aware Interaction - Context Model **5**

7 User Journey **5**

8 Monitoring Interface **6**

9 AI Integration **7**

10 Why HollowHaven is a Ubicomp System **7**

11 Current Prototype Capabilities **8**

12 Current Limitations **8**

13 Evaluation Plan **9**

14 Occupancy and audio evaluation **10**

15 Future Development **10**

16 AI Acknowledgement **11**

1 Project Overview

1.1 Background

Greater Glider is an endangered native Australian species whose habitat is rapidly declining. The loss is driven by many factors including irreversible land clearing and logging, devastating bushfires. Because natural hollows take many decades to form, and because greater gliders depend strongly on mature, old-growth forests that provide suitable tree hollows for shelter and breeding, the loss of old trees creates a serious shortage of suitable habitat (WWF-Australia, n.d.). Artificial tree hollows therefore provide a practical way to supplement declining natural nesting and shelter sites.

1.2 Description

Our project title is: **HollowHaven: A Context-Aware Artificial Tree Hollow for Greater Gliders.**

This is device is an artificial tree hollow designed to reduce the overheating risk for Greater Glider. The current prototype combines environmental sensing, a controllable water-delivery mechanism, occupancy-event detection, local data logging, and optional AI-assisted audio analysis.

The system is built around an ESP32 microcontroller connected to environmental sensors, a servolocal Node.js service running on a computer receives telemetry and recorded audio from the ESP32 and presents the information through a browser-based monitoring interface.

1.3 Product Vision

Artificial tree hollows are increasingly being used to compensate for habitat loss. But unlike the mature natural tree hollows, the temperature buffering capacity of artificial one is limited. HollowHaven transforms traditional tree holes into an adaptive ones that can sense environmental conditions, respond to overheating risks and provide useful information for wildlife caregivers and researchers.

2 Problem and Design Opportunity

2.1 Ecological Problem

Greater gliders rely on mature tree hollows for shelter, breeding and protection from extreme environment. However, the formation of natural tree holes may take decades, while the speed of habitats loss and tree removal is slower than formation.

Artificial hollows are therefore increasingly used as habitat substitutes. But their thermal properties may differ significantly from natural hollows.

2.2 Core Problem: Overheating

The main challenge that HollowHaven addresses is overheating in artficial hollows during hot weather. The traditional artificial hollows primarily provide physical shelter but usually have little ability to sense or actively respond to changing environmental conditions. Midday during the day is generally when the temperature is highest. The temperature in most artificial tree holes exceeds the suitable temperature for greater gliders to live in natural tree holes at this time.

3 Design Goals

Goal 1: Reduce overheating risk
Goal 2: Respond automatically to environmental changes
Goal 3: Preserve a naturalistic habitat
Goal 4: Support monitoring and maintenance. The staff can know current temperature, humidity, water level, whether the cooling system is working and whether water needs to be added or maintenance is required.

4 Proposed Solution

4.1 Environmental Monitoring

The system continuously monitors the internal environment through multiple sensors:

- Internal temperature and relative humidity;
- Moisture conditions of the water-guiding material;
- Ambient light intensity;
- Water-level status;
- Acoustic activity.

These measurements allow the system to distinguish different aspects of the artificial hollow environment instead of relying on a single sensor.

4.2 Water-Delivery Control

The current prototype uses light intensity as the direct control input for the water valve. When the light sensor detects a value below the configured threshold, the ESP32 requests the servo-driven valve to open.

The valve therefore represents the actuation component of the system: contextual information collected from the environment can result in a physical change to the habitat.

Importantly, the current prototype does not use temperature or humidity as direct valve-control inputs. These values are retained for environmental observation and future evaluation.

4.3 Water-Delivery Verification and Maintenance

HollowHaven does not treat a valve-open command as proof that water has successfully reached the intended cooling material.

Instead, the system monitors two moisture-sensing points on the water-guiding layer. If the valve has remained fully open for at least 60 seconds and either monitored position is still classified as dry, the system generates a possible outlet-blockage maintenance warning.

A separate water-level sensor provides an independent refill warning. If the water in the tank is below a certain value, the backend will receive refill warning. This means that the system distinguishes between: a low-water condition, and a possible unsuccessful water-delivery condition.

4.4 Visitor Observation

A VL53L0X distance sensor monitors whether a nearby object is present in the entrance area. A confirmed transition from an unoccupied state to a suspected occupied state generates an entry event.

The event triggers an approximately 30-second audio recording. The audio is then transferred to the local computer, stored as a WAV file, and made available through the monitoring interface.

Users can manually submit a recording to an AI-based sound-analysis service. The returned result is displayed only as an AI candidate requiring verification, rather than as confirmed species identification.

5 System Architecture

5.1 Sensor Basic Information

HollowHaven follows a distributed local architecture consisting of an embedded sensing device and a computer-based monitoring service.

The ESP32 acts as the central embedded controller. It collects environmental measurements, evaluates local control rules, drives the servo valve, detects visitor events, and records audio.

The computer and ESP32 operate on the same local network. The ESP32 uses 2.4 GHz Wi-Fi in station mode rather than creating its own hotspot.

Communication is separated into three channels:

- **UDP port 9002**: device discovery;
- **TCP port 9000**: telemetry transmission;
- **TCP port 9001**: audio transmission.

A local Node.js service receives the telemetry and audio. The browser does not communicate directly with the ESP32. Instead, it connects to the Node.js service through a locally hosted web interface at port 8766.

The computer stores environmental records as CSV files and audio as WAV files. Optional AI sound analysis is initiated manually through the web interface.

Architecture Flow: Sensors → ESP32 → Local control and event detection → Wi-Fi communication → Node.js local service → Local storage and browser interface → Optional manual AI analysis

5.2 Sensor and Actuator Design

The sensor suite is intentionally heterogeneous because each sensor answers a different question.

Temperature and humidity describe the internal environment; the moisture sensors indicate whether water reaches the intended material; the water-level sensor indicates whether maintenance is required; the distance sensor observes possible entry events; and the microphone provides additional evidence after an entry event.

Table 1: Components and their functions in HollowHaven

Component	Function in HollowHaven
SHT3x	Measures internal temperature and relative humidity.
Capacitive moisture sensor 1	Monitors moisture at the first position of the water-guiding material.
Capacitive moisture sensor 2	Monitors moisture at a second position and contributes to abnormal watering detection.
Light sensor	Current direct input for valve-open/close decisions.
Water-level sensor	Provides refill-maintenance warning.
VL53L0X	Detects nearby objects and generates suspected occupancy events.
SEN0487 microphone	Measures acoustic activity and records event audio.
MG90S servo valve	Physically opens or closes the water path.

6 Context-Aware Interaction - Context Model

HollowHaven differs from a conventional screen-based interactive system because its primary interactions are implicit.

The greater glider is not expected to actively operate buttons, menus, or other controls. Instead, environmental and physical events become inputs to the system.

The current prototype recognises several contextual conditions:

Direct sunlight condition
If the valid light ADC value is below 500, the system requests the valve to open.

Water-delivery condition
The system observes whether the water-guiding material becomes wet after the valve reaches the open position.

Maintenance condition
If the water-level reading is below the configured threshold, the system generates a refill warning.
If the valve has been fully open for at least 60 seconds while either moisture sensor still reports a dry state, the system generates a possible blockage warning.

Suspected occupancy condition
When the system is currently idle and receives three consecutive valid distance measurements of 189 mm or less, it changes to a suspected occupied state and triggers an entry event.
Five consecutive valid measurements of 220 mm or more are required to return from occupied to idle.
This design uses hysteresis and repeated confirmation in selected sensing tasks to reduce unstable state transitions.

7 User Journey

HollowHaven supports two different types of users:

- the **biological occupant**, such as a greater glider;

- the **human operator**, such as a researcher, wildlife carer, or project maintainer.

Step 1 — Deployment
The operator installs the artificial hollow, connects the sensors and water-delivery system, fills the reservoir, powers the ESP32, and connects the computer and ESP32 to the same local network.

Step 2 — Passive environmental observation
During normal operation, the system continuously collects temperature, humidity, moisture, light, water-level, distance, and sound-related measurements.
The biological occupant does not need to intentionally interact with the system.

Step 3 — Water-delivery response
When the configured light condition is detected, the ESP32 requests the servo valve to open.
The valve moves gradually to the target position.

Step 4 — Water-delivery verification
Once the valve is fully open, the system begins monitoring how long the valve remains open and whether the water-guiding material becomes wet.
If either monitored area remains dry for at least one minute after the valve is fully open, a maintenance warning is produced.

Step 5 — Maintenance support
The water-level sensor independently checks whether the reservoir requires refilling.
The operator can inspect refill and blockage warnings from the browser interface.

Step 6 — Visitor event
When a new nearby object is confirmed by the distance sensor, the system generates an entry event and begins an approximately 30-second audio recording.

Step 7 — Audio review
The computer receives and saves the recording.
The operator can: play the recording; inspect the waveform; check recording information; download the file; optionally submit it for AI analysis.

Step 8 — Human interpretation
The AI output is displayed as a candidate result rather than as a confirmed identification.
Final interpretation and maintenance decisions remain with the human user.

8 Monitoring Interface

The browser-based interface provides a central view of the current HollowHaven state.

The main monitoring page displays: device connection status; reservoir refill condition; sunlight condition; valve state; moisture state at both monitored locations; maintenance warnings; suspected occupancy state; recording state; temperature; relative humidity; distance; environmental trends; sound RMS and peak-to-peak values; observation events.

The current backend stores up to 900 recent telemetry records in memory, corresponding to approximately 30 minutes at the normal telemetry interval.

Recorded audio is presented separately in the sound archive, where users can inspect, play, and analyse individual WAV recordings.

The interface is designed primarily as a monitoring and maintenance tool rather than as a direct remote-control panel. For example, the current browser interface does not provide a manual valve-open/close button.

9 AI Integration

AI is integrated into HollowHaven as an optional interpretation layer rather than as an autonomous decision-making controller.

When a visitor event generates an audio recording, the recording is stored locally first. No recording is automatically uploaded to an external AI service.

The operator may manually select a WAV file and submit it either to Gemini audio understanding or to a configured custom classification service.

The returned analysis may contain: a candidate label; supporting acoustic evidence; uncertainty information; model or service information.

The interface explicitly presents the result as an **AI candidate requiring verification**.

This separation is important because the audio system does not provide reliable species confirmation, and the current microphone and sampling configuration are not designed to capture the complete ultrasonic vocalisations of bats.

AI results therefore support human interpretation but do not: confirm animal identity; change occupancy state; control the valve; trigger autonomous ecological intervention.

10 Why HollowHaven is a Ubicomp System

HollowHaven demonstrates several core characteristics of ubiquitous computing.

Embedded computing: The computing hardware is embedded into a physical wildlife shelter rather than existing as a conventional desktop application.

Context awareness: The system observes environmental and physical context including light, moisture, water level, distance, temperature, humidity, and acoustic activity.

Implicit interaction: The biological occupant does not explicitly operate the system. Environmental changes and physical presence become interaction inputs.

Physical actuation: The system can translate sensed context into physical action through the servo-controlled valve.

Distributed interaction: Sensing and actuation occur on the ESP32, while storage, visualisation, and optional AI analysis occur on a connected computer.

Calm interaction: Most sensing and control activities occur in the background. Human attention is mainly required when maintenance, interpretation, or inspection is necessary.

11 Current Prototype Capabilities

The current HollowHaven v4 prototype has implemented:

- internal temperature and humidity monitoring;
- dual-point moisture monitoring;maintainer.
- light-based valve-control requests;
- gradual servo-driven valve actuation;
- water-level maintenance warnings;
- possible water-delivery blockage warnings;
- distance-based suspected occupancy detection;
- event-triggered approximately 30-second audio recording;
- local telemetry storage;
- local WAV storage;
- browser-based monitoring;
- optional manually triggered AI audio analysis.
- These functions demonstrate a complete sensing–communication–interpretation–actuation pipeline, while still keeping critical ecological decisions under human supervision.

These functions demonstrate a complete sensing–communication–interpretation–actuation pipeline, while still keeping critical ecological decisions under human supervision.

12 Current Limitations

The current prototype should be interpreted as a functional proof of concept rather than a fully validated ecological product.

Several limitations remain.

First, temperature and humidity are currently observational variables and are not used in a closed-loop thermal-control algorithm.

Second, the valve state is inferred from the commanded PWM position. There is no mechanical valve-position feedback and no direct water-flow sensor. Therefore, a displayed "open" state does not prove that water is physically flowing.

Third, a low-water condition currently generates a warning but does not automatically prevent the system from requesting the valve to open.

Fourth, the moisture sensors provide raw ADC-based dry/wet classifications rather than calibrated water-content measurements.

Fifth, the distance sensor detects nearby objects rather than species identity. "Suspected occupancy" therefore does not prove that a greater glider is inside the hollow.

Sixth, AI audio analysis provides candidate interpretation only and does not provide trusted species confirmation.

Finally, the effectiveness of evaporative cooling, the long-term outdoor reliability of the system, and its ecological suitability still require physical and field validation.

13 Evaluation Plan

The evaluation of HollowHaven should be separated into several dimensions rather than relying only on interface status.

13.1 Functional Evaluation

Verify whether:

- sensor measurements are successfully collected;
- the light threshold generates the correct valve request;
- servo movement reaches the expected target positions;
- moisture-state hysteresis behaves correctly;
- water-level warnings appear at the expected threshold;
- the blockage warning is generated after the required open-valve duration;
- entry and exit confirmation rules behave correctly;
- audio recording, transfer, and storage complete successfully.

13.2 Environmental Evaluation

Compare environmental measurements before, during, and after water delivery.

Important metrics may include:

- internal temperature;

- relative humidity;
- maximum temperature;
- temperature change after watering;
- time required for environmental change.

This evaluation is necessary before claiming that HollowHaven provides effective cooling.

13.3 Water-delivery evaluation

Physically verify:

- whether the valve actually opens;
- whether water flows through the intended path;
- whether both monitored areas become wet;
- whether blockage warnings correspond to real delivery failures.

14 Occupancy and audio evaluation

Test whether:

- nearby-object detection is stable;
- entry events are generated only after the intended confirmation period;
- recordings are complete;
- AI candidates are useful but not overinterpreted as confirmed identification.

15 Future Development

Future versions of HollowHaven could extend the current prototype in several directions.

The most important next step is to introduce validated thermal-control logic. After sufficient environmental testing, temperature, humidity, and possibly external temperature could be incorporated into the decision process rather than relying only on light intensity.

Additional improvements could include:

- direct water-flow sensing;
- mechanical valve-position feedback;
- automatic low-water protection;
- calibrated moisture measurements;
- outdoor waterproofing and long-term power management;
- improved species-detection methods;

- remote monitoring of multiple hollow units;
- long-term field deployment and ecological evaluation.

The long-term goal is to develop HollowHaven from a monitored prototype into a reliable adaptive artificial habitat that can respond to environmental stress while still maintaining transparent and human-verifiable control logic.

16 AI Acknowledgement

Generative AI was used during the preparation of this project report. The AI tool used was ChatGPT (GPT-5.6 Sol), developed and published by OpenAI, available at https://chatgpt.com/.ChatGPT was used primarily to support brainstorming, outlining, paraphrasing, and language refinement. In particular, it assisted in improving the clarity and structure of written descriptions related to system requirements, use cases, architecture design, and UML modelling. It was also used to review wording, grammar, and formatting suggestions for the report.